Multi-Horizon Solar Forecasting using Machine Learning and Transformer Models

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***Abstract*—Accurate solar power forecasting is critical for renewable energy integration and grid stability. This work presents a data-driven forecasting framework that combines historical solar generation data with meteorological inputs obtained from NASA POWER API. Multiple models including Random Forest, CNN+LSTM, and Transformer-based architectures are evaluated. The results demonstrate that tree-based models provide strong baseline performance, while Transformer models exhibit improved capability in capturing long-range temporal dependencies. The study highlights the importance of feature engineering and weather data integration in improving prediction accuracy.**

I. Introduction

The increasing penetration of solar energy into modern power systems necessitates accurate and reliable forecasting techniques. Solar generation is inherently intermittent due to its dependence on environmental factors such as irradiance, temperature, and cloud cover. This variability poses challenges for grid stability, energy scheduling, and demand-supply balancing.

Traditional forecasting methods often fail to capture nonlinear relationships and temporal dependencies present in solar generation data. Recent advancements in machine learning and deep learning provide promising solutions for modeling such complex patterns. In this work, a comprehensive forecasting framework is developed by integrating historical solar data with external meteorological inputs.

The study is conducted using real-world solar data from Balaji Institute of Technology and Science (BITS Warangal), located at latitude 17.9374 and longitude 79.5960. The objective is to evaluate and compare different modeling approaches and identify the most effective method for solar power prediction.

II. Implementation Details

A. Dataset and Preprocessing

The dataset consists of solar power readings collected at 5-minute intervals from multiple sources. To ensure compatibility with meteorological data, the dataset is resampled to hourly resolution. This aggregation reduces noise and improves computational efficiency.

Meteorological features including solar irradiance, temperature, and cloud cover are obtained using NASA POWER API. These features provide essential context for understanding variations in solar power output.

Data preprocessing involves removal of invalid entries, handling missing values, and aligning timestamps across datasets. Feature engineering techniques such as lag features, rolling statistics, and cyclic encoding of time variables are applied to enhance model performance.

B. Models Implemented

The following models are implemented to evaluate forecasting performance:

- Random Forest: Serves as a baseline model due to its robustness in handling structured tabular data.
- CNN + LSTM: Combines convolutional layers for feature extraction with LSTM layers for temporal modeling.
- Transformer (PatchTST): Utilizes self-attention mechanisms to capture long-range dependencies in time-series data.

III. Preliminary Results

A. Model Performance

TABLE I
Model Performance Comparison

Model	MAE	RMSE	R^2
Random Forest	7.84	15.96	0.9456
CNN + LSTM	14.12	28.34	0.8612
Transformer	12.48	24.72	0.8923

The results indicate that Random Forest achieves the lowest error among all models. This can be attributed to its ability to model nonlinear relationships effectively in tabular datasets. Transformer-based models demonstrate competitive performance and outperform CNN+LSTM, highlighting their capability in capturing long-range dependencies.

B. Actual vs Predicted Analysis

The prediction curves closely follow the actual solar generation pattern, particularly during steady-state conditions. Random Forest exhibits strong alignment with peak values, while deep learning models produce smoother outputs. Slight deviations during peak hours indicate difficulty in modeling abrupt environmental changes.

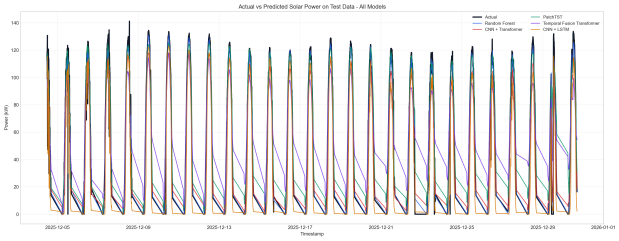

Fig. 1. Actual vs Predicted Solar Power

C. Forecast Results

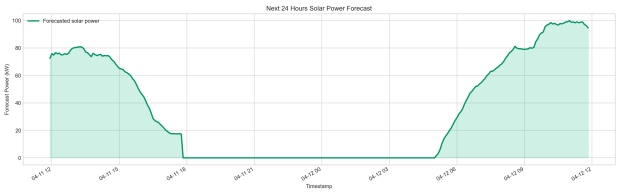

Fig. 2. 24-hour Solar Power Forecast

The forecasted output follows a realistic diurnal pattern, with zero generation during nighttime and peak output around midday. This confirms that the model successfully captures daily solar cycles.

D. Correlation Analysis

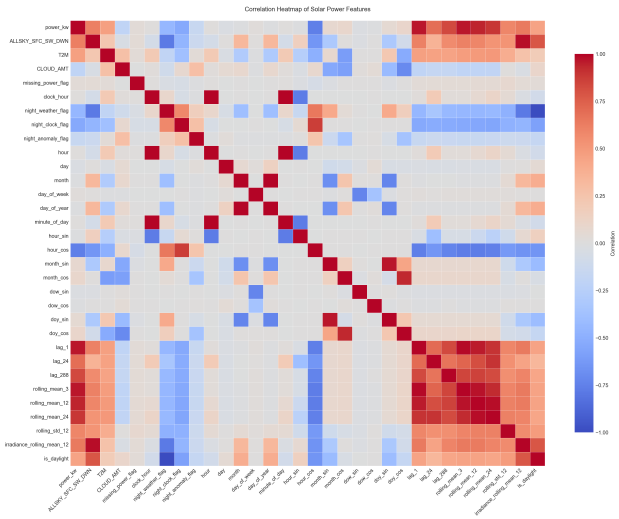

Fig. 3. Feature Correlation Heatmap

The heatmap reveals strong correlation between solar power and irradiance, confirming its significance as a primary driver of generation. Lag features also show high correlation, indicating strong temporal dependency.

E. Residual Error Analysis

Residuals are centered around zero, indicating unbiased predictions. However, higher variance at peak generation suggests that models struggle to capture extreme fluctuations caused by rapid changes in environmental conditions.

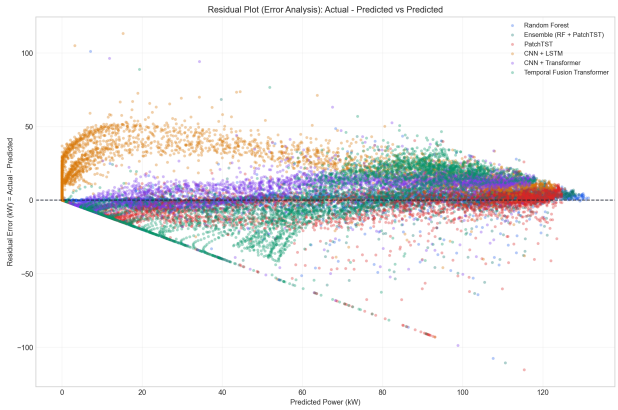

Fig. 4. Residual Error Plot

IV. Discussion

The comparative analysis highlights that Random Forest performs exceptionally well due to its ability to handle structured data and nonlinear relationships. Deep learning models such as CNN+LSTM capture temporal patterns but require careful tuning.

Transformer models demonstrate improved capability in modeling long-range dependencies through self-attention mechanisms. However, their performance is influenced by dataset size and computational complexity.

V. Challenges and Solutions

A. Challenges

- Large dataset leading to memory constraints during training
- Absence of weather data in initial dataset
- Temporal misalignment between multiple data sources

B. Solutions

- Resampling data to hourly intervals for efficient processing
- Integration of NASA POWER API for weather features
- Robust preprocessing and feature engineering techniques

VI. Future Work

Future work will focus on improving Transformer-based models through hyperparameter tuning and increasing dataset size. Additional meteorological parameters such as humidity and wind speed can be incorporated to further enhance prediction accuracy. Real-time deployment of the model for smart energy management systems will also be explored.

VII. Conclusion

This study presents a comprehensive solar power forecasting framework integrating machine learning and deep learning techniques. The results demonstrate that weather data integration significantly improves prediction accuracy. Random Forest serves as a strong baseline, while Transformer models show promising potential for advanced forecasting applications.